

Load Interconnection Request System Impact Study Report

For Queue Project Q17-0743

Christopher 100 kV
October 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0743. The Interconnection Customer proposes a 63.1 MW large load (data center) facility interconnecting at the Christopher 100 kV substation in the Indiana area. The request was received on February 09, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 268 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,199,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lower Granite-Christopher 100kV	Murdock-Napavine 100kV	128.4	136	83.3
Canal St-Lowe Granite 100kV	Napavine-Murdock Substation 100kV	107.7	188	83.3
Red Lion-St. Rita 100kV L	Murdock 230/1kV Autotransformer	105.4	166	30.1
S Haven-Napavine 100kV Line	Canal St Substation-Lower Granite Substation 100kV	105.0	153	62.3
Napavine-Murdock 100kV	Lower Granite-Canal St 100kV	103.6	166	62.3
Desoto-Federalgen Tap 100kV line	Murdock 230-1 kV TX	103.0	251	40.8
Mulberry Cogeneration Facility-St. Rita 100kV	Murdock 230 kV/1 kV Autotransformer	102.2	162	30.1
Christopher-Rmec 100kV L	Lower Granite-Canal St 100kV Line	100.7	124	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lower Granite-Christopher 100kV	Rebuild line with higher-capacity conductor	\$1,706,000	21% (\$357,000)
Canal St-Lowe Granite 100kV	Reconductor line	\$733,000	21% (\$153,000)
Red Lion-St. Rita Substation 100kV	Reconductor line	\$1,180,000	19% (\$229,000)
South Haven-Napavine 100kV	Reconductor line	\$2,247,000	17% (\$374,000)
Napavine-Murdock 100kV L	Reconductor line	\$3,081,000	17% (\$512,000)
Desoto-Federalgen Tap 100kV line	Reconductor line	\$732,000	19% (\$136,000)
Mulberry Cogeneration Facility-St. Ita Substation 100kV	Reconductor line	\$1,367,000	19% (\$265,000)
Christopher-Rmec 100kV line	Reconductor line	\$2,173,000	100% (\$2,173,000)
	Total allocated to project		\$4,199,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.